NORMALITY OF A LOCAL RING, OF ITS HENSELIZATION AND OF ITS STRICT HENSELIZATION ARE EQUIVALENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a local ring, normal means normal domain. Algebra, Definition \ref{algebra-definition-ring-normal}, cited at proof lines 11119-11121. Reconciles the two formulations used in the statement.
- Normality descends along faithfully flat ring maps. Algebra, Lemma \ref{algebra-lemma-descent-normal}, cited at proof lines 11123-11124. Gives one direction of the equivalence.
- Normality ascends along smooth ring maps and is preserved by filtered colimits. Algebra, Lemma \ref{algebra-lemma-normal-goes-up} and Algebra, Lemma \ref{algebra-lemma-colimit-normal-ring}, cited at proof lines 11127-11129. Give the converse direction from the colimit presentation by etale algebras.

Source reference. Original lemma 06DI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8182–8182. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00GV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 621–621. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4055–4055. Navigation pointer to a statement the

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Source reference. Original lemma 033C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4051–4051. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 037D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4196–4196. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a local ring. The following are equivalent: R is a normal domain, the henselization R^h of R is a normal domain, and the strict henselization R^{sh} of R is a normal domain.*

Proof. A preliminary remark is that a local ring is normal if and only if it is a normal domain (see Algebra, Definition 00GV). The ring maps $R \to R^h \to R^{sh}$ are faithfully flat. Hence one direction of the implications follows from Algebra, Lemma 033G. Conversely, assume R is normal. Since R^h and R^{sh} are filtered colimits of étale hence smooth R-algebras, the result follows from Algebra, Lemmas 033C and 037D. □